



TARGET POPULATION

ALL ADULTS ≥ 18 YEARS OF AGE

HIGHER RISK POPULATION

- Long standing hypertension, diabetes mellitus, chronic kidney disease, arrhythmia (Atrial Fibrillation)
- Structural heart disease: Valvular heart disease (VHD) including Rheumatic heart disease (RHD), CAD (Coronary Artery Disease), DCM (Dilated Cardiomyopathy)
- Family history of heart disease
- Obesity and/or Sleep apnea
- Chronic alcohol consumption



CLINICAL DIAGNOSIS

FRAMINGHAM CRITERIA

MAJOR CRITERIA

- Dyspnea that will awake patient after 2-3 hours of sleep
- Dyspnea on lying down/ Difficulty lying flat
- Weight loss of 4.5 kg or more in 5 days due to diuretic therapy
- Neck vein distention
- Crackles (Rales): Abnormal lung sounds
- Third heart sound heard with a stethoscope
- Enlarged heart, visible on a chest X-ray (If available)

MINOR CRITERIA

- Dyspnea with ordinary physical activity (walking, climbing stairs e.t.c)
- Swelling in the legs, ankles, or abdomen
- Persistent cough
- Heart rate > 120 beats per minute at rest
- Enlarged liver
- Pleural effusion

DIAGNOSIS:

Heart Failure is likely If: 2 major criteria or 1 major and 2 minor criteria are met

Disclaimer: This leaflet is currently being piloted and may contain inconsistencies. Please help us improve it by reporting any issues to KHDC Nepal.

DRUG LIST AND DOSE

Class: Diuretics Drug: Furosemide** 20 mg-40 mg	Side effects Hyponatremia, Hypokalemia, Hypovolemia
Class: Beta-blockers Drug: Metoprolol succinate** 12.5-50 mg	Side effects Bronchospasm, Bradycardia
Class: RAAS Inhibitors (ACEi/ARB) Drug: Enalapril** 5-10 mg / Losartan** 25-50 mg	Side effects Hyperkalemia, Cough, Angioedema
Class: SGLT2 inhibitors Drug: Empagliflozin* 10 mg	Side effects Genitourinary infection, Vulval candidiasis
Class: MRAs (Mineralocorticoid Receptor Antagonists) Drug: Spironolactone* 25-50 mg	Side effects Gynaecomastia, Hyperkalemia

*National list of essential medicines, 6th revision, MOHP Nepal

**Health insurance board benefit package, January 2024

CONGESTIVE HEART FAILURE - "The Donkey" analogy



DECOMPENSATED CHF: Cardiac output not able to keep up with demand. Preload increased due to fluid and sodium retention. Afterload increased due to sympathetic nervous system activation and vasoconstriction.



ACE INHIBITORS/ARBs: Reduces afterload / preload by blocking angiotensin/aldosterone.
FUROSEMIDE: Reduces preload by decreasing fluid and sodium retention.
HYDRALAZINE/NITRATES: Reduces afterload/preload by blocking vasoconstriction/vonodilation.



BETA-BLOCKERS: Inhibit negative remodeling from activated sympathetic nervous system.



INOTROPES (DIGOXIN, DOBUTAMINE, MILRINONE): Increase contractility and cardiac output.

Ref: Learntheheart.com

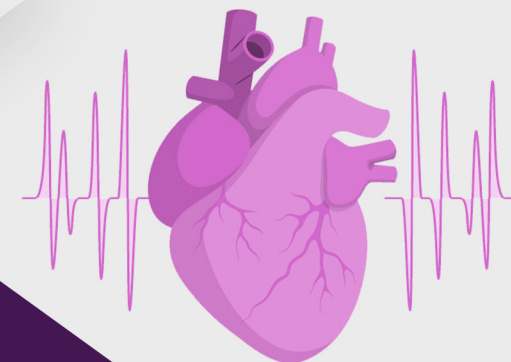
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HEART FAILURE

IDENTIFICATION AND MANAGEMENT FOR PRIMARY CARE PROVIDERS

USE THIS GUIDE TO

- Identify patients with Heart Failure
- Detect Heart Failure among patient at risk
- Manage and refer patients with Heart Failure

Heart failure (HF) is a condition in which the heart does not pump blood effectively to meet the needs of the tissue for oxygen and nutrients.



CLINICAL FEATURES



Shortness of breath



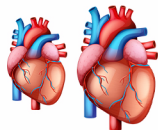
Palpitation



Fluid surrounding the lungs



Weakness



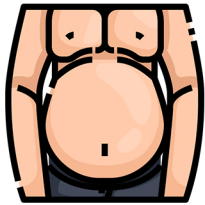
Enlarged Heart



Swollen feet

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**THE FLUID IN THE BODY STARTS TO ACCUMULATE
(JUST LIKE BACK LOG IN A BLOCKED DRAIN).**



Ascites

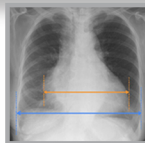
Fluid buildup in abdomen (bloated belly)



Orthopnea

Difficulty breathing while lying down

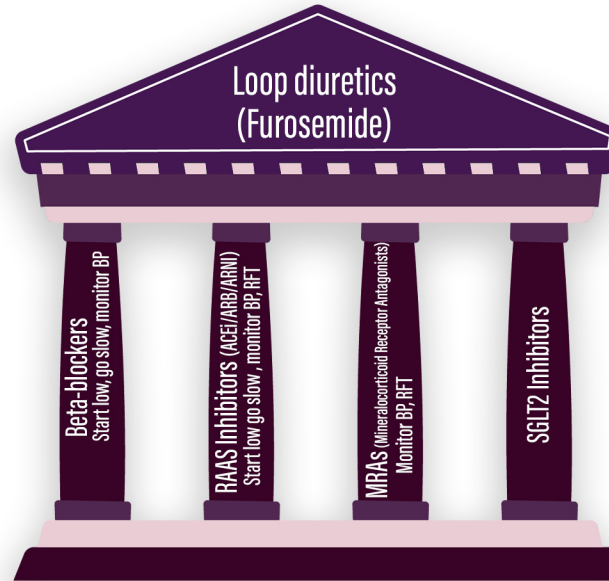
INVESTIGATIONS (*If available)



Clinical diagnosis is confirmed by **Echocardiography***
Electrocardiogram(ECG)* for any abnormal heart rhythm, **Hemoglobin*** for anemia, **Kidney function Test ***



PHARMACOLOGICAL TREATMENT



Titrate to highest tolerated dose

- Start Furosemide at 40 mg PO OD and if dyspnea persists then Refer
- Beta blocker dose has to target pulse rate between 50-60 bpm

Note: Beta blocker is contraindicated in acute heart failure



NON-PHARMACOLOGICAL ADVICE



Regular Physical Activity



Reduce Stress



Low Salt Consumption



Quit Alcohol & Tobacco



Consume Healthy Diet



Weight Reduction



Adequate Sleep

Note: All heart failure patients should reduce fluid intake less than 1.5 to 2 litre



CRITERIA FOR REFERRAL

- All newly diagnosed or suspected patients with heart failure
- Worsening dyspnea, edema or fatigue
- Presence of jaundice
- Abnormal heart rhythm
- Bradycardia and/or Tachycardia
- Chest pain (after initial stabilisation using KHDC leaflet for Ischemic heart disease)
- NYHA (New York Heart Association) Class III (dyspnea in less than ordinary activity) and IV (dyspnea at rest)
- Dyspnea not relieved even after Furosemide dose of 40 mg

**SEEK TELEMEDICINE HELP
FOR ANY CLINICAL QUERIES**

HEART FAILURE IS THE MAJOR CONTRIBUTOR TO DEATH, DISABILITY AND RISING HEALTHCARE COST.

- Common causes of Heart failure in Nepal are:
- Valvular heart disease including Rheumatic heart disease
 - Hypertension, Dilated cardiomyopathy
 - Coronary artery disease

ALSO REFER TO KHDC CVD RISK ASSESSMENT

CONTACT US

For any clinical queries / If anything in the document is difficult to understand



+977-9852072088 / 9802766889

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